BINANCE RESEARCH

A Primer on On-Chain Gaming

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1 Key Takeaways

- On-chain gaming stands apart from traditional and hybrid gaming models by having its entire game logic and state hosted directly on the blockchain.
- A fully on-chain gaming model offers the advantages of immutability and data permanence; composability and interoperability; community governance and true asset ownership. As growth in blockchain gaming is expected to witness a compounding annual growth rate ("CAGR") of 70.3% till 2025, fully on-chain games will likely see increased adoption as well. Despite these benefits, this emerging sector has to overcome several challenges such as technical constraints, poor user experience, bot interferences, and a fragmented gaming ecosystem.
- Key contributors to the development of fully on-chain games include infrastructure providers, gaming studios, and the users themselves, each playing a pivotal role in realizing this vision.
- On-chain games have garnered significant interest thus far as we saw notable game developers such as Proof of Play and Neon Machine raising US$33M and US$20M respectively to develop on-chain games. Progress and interest in on-chain games have also grown in tandem with gaming titles such as Nine Chronicles rising through the ranks to become the top 10 most popular games on DappRadar.
- Though not imminent, the future of fully on-chain games is well-supported, with three key catalysts acting as tailwinds for mass adoption of on-chain gaming: engaging gameplay with sustainable design, advancements in gaming infrastructure and finally, the development of a more interoperable game economy, which is often referred to as the 'Autonomous World'.

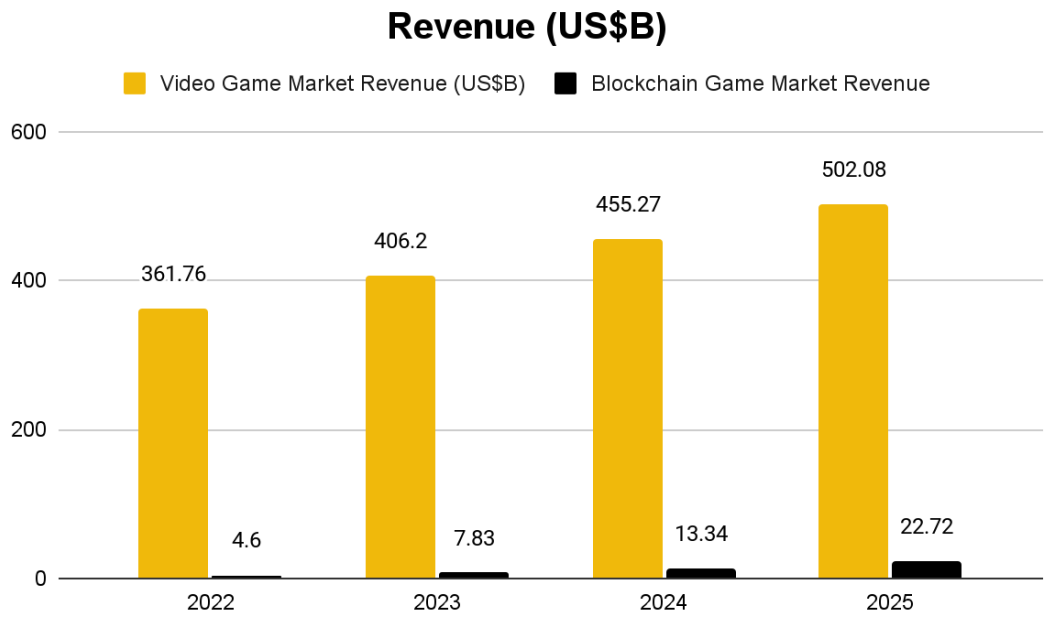
2 Background

2.1 Traditional vs On-Chain Games

Blockchain games first started in 2014 with the launch of Huntercoin to test the potential of integrating blockchain technology with games. Even though blockchain games promise to deliver benefits such as digital asset ownership, and community governance, the ecosystem for blockchain games barely scrapes the tip of the iceberg when compared to traditional games. **For a sense of scale, Alien World, one of the most popular blockchain games, is less than 0.1% the size of Minecraft in terms of average monthly active users. That is around 1 player on Alien World to 932 players on Minecraft.** Evidently, the adoption of blockchain technology is still in its infancy with fully on-chain games only accounting for about 5% of all blockchain games[(1)].

To date, the traditional video game industry holds the **largest gaming community** and is a lucrative industry with market revenue expected to see **a CAGR of 11.2% from 2023 to 2025[(2)]**. With the size of the blockchain gaming market expected to record a **CAGR of 70.3% and cross US$22B by 2025[(3)]**, fully on-chain games being a subset of blockchain games are likely to see a similar growth trajectory.

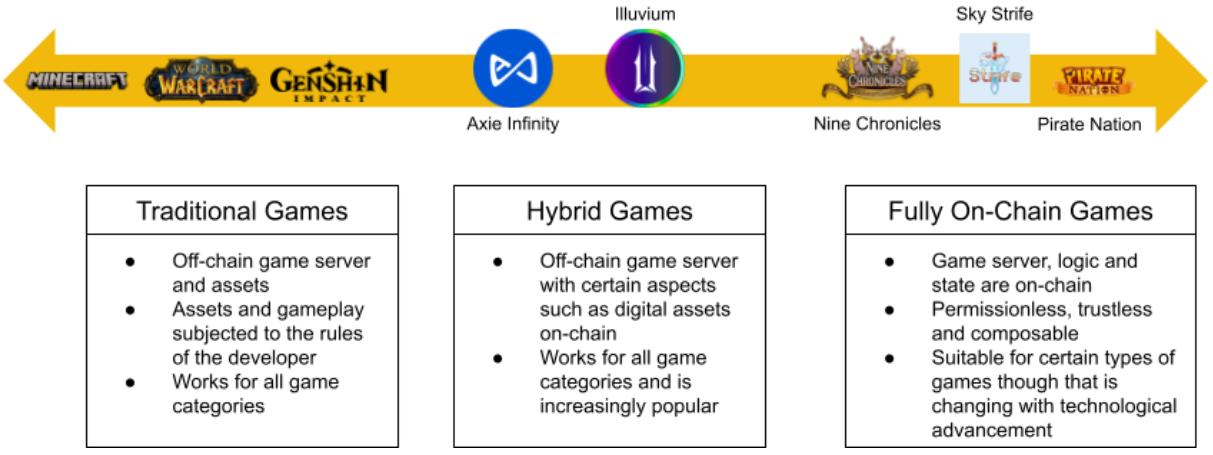
Figure 1: Size of Traditional Video Game Market



Source: Statista, Binance Research

However, there are exciting innovations happening in the space and we have seen a shift from the traditional gaming model towards what is known as **"fully on-chain games"**. **Traditional games are games where the logic and game state are stored on a centralized, off-chain server.** For context, **blockchain games refers to games that leverage blockchain technology for its in-game assets, although game server and logic are off-chain and centralized.** This exposes users to platform risk should the game shut down. However, **in fully on-chain games, game logic and game state are all stored on-chain, existing forever as long as the underlying blockchain technology is still running.** Game logic refers to the rules and mechanics that define how the game operates, such as transfers, movements, combat etc. Game state records the current status of the game and is updated regularly to reflect changes in the gaming environment such as when there is an action taken, change in ranks etc.

Figure 2: Game spectrum with increasing appetite for on-chain gaming



Source: Binance Research

According to Brian Gu, founder of Dark Forest which is one of the earliest implementations of a fully on-chain game, a fully on-chain game should possess the following characteristics[4]:

- The source of truth lies in the blockchain which is used as a shared state for all game updates.
- Game logic and rules are implemented via smart contracts.
- Game is based on open ecosystem principles.
- Game is client agnostic.
- Games are interoperable with things we consider valuable.

In this report, we will only examine the state of fully on-chain games and its potential to capture a meaningful share of the gaming industry.

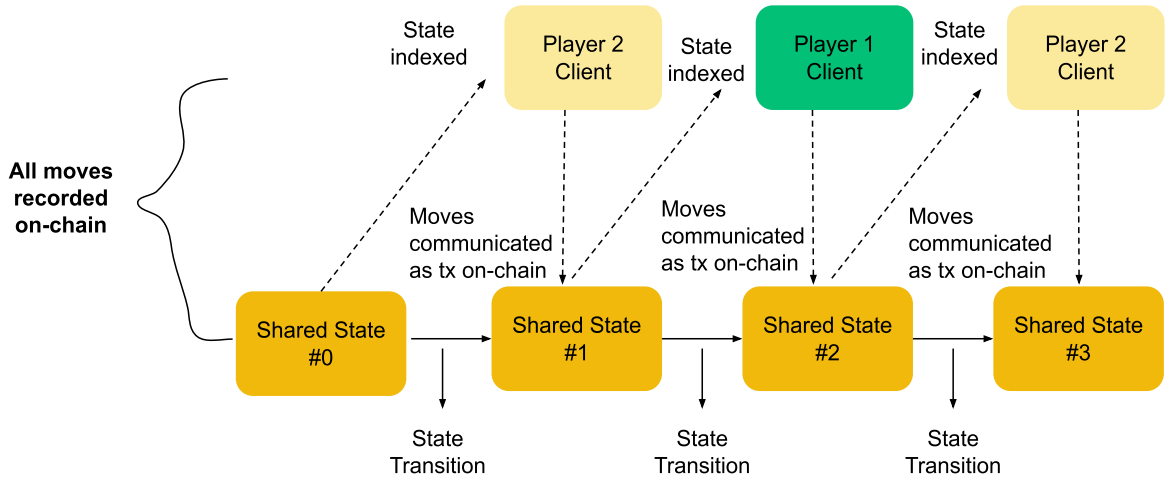
2.2 Value Proposition

Fully on-chain games are desirable due to value propositions such as **immutability and game permanence, composability and interoperability, true ownership of digital assets and community governance**.

Immutability and Game Permanence

The move towards fully on-chain games aims to capture the fundamental benefits of blockchain technology: immutability and game permanence. In traditional gaming, if the gaming publisher goes bankrupt or sunsets the game, all assets of the game would disappear too. Nintendo's closure of its eShop for example has put many traditional classic games now out of reach. Contrast it against a game that lies entirely on the blockchain, as long as the network continues to operate, the game will continue to exist. This offers a **persistence guarantee that significantly derisks the invested resources spent** by gamers in creating assets and driving the development of the gaming ecosystem. The cryptographic and transparent nature of blockchain also prevents game data from being tampered with.

Figure 3: On-Chain games will continue to exist as long as the blockchain lasts



Source: Jump Crypto, Binance Research

Composability and Interoperability

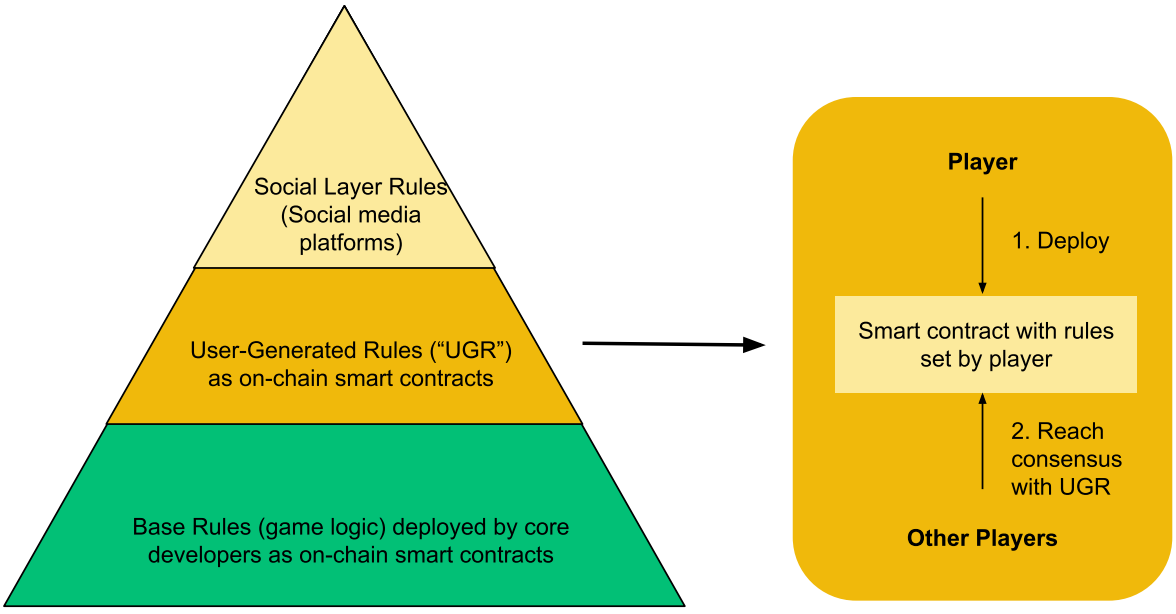
One of the greatest benefits of on-chain gaming is permissionless composability given to gamers and developers. **Instead of operating within siloed environments controlled by most traditional game publishers, on-chain games allow users to take the code, items, assets from one application and use them in another.**

The desire to create new content is not unique to on-chain games. Modding has long been a key component in web2 games, with players showing a key desire to improve upon existing

game concepts. According to Mod.io[5], the popularity of user-generated content ("UGC") games is expected to grow exponentially and is predicted to consume $1 for every $10 spent in video games by 2025. With games like LoL, DoTA, and PUBG having their games built on modded versions of other games, it is apparent that modding is an attractive and important element in games. Roblox, one of the largest community-driven gaming platforms, has hosted more than 40 million user-generated games[6] and also recorded a 20% YoY growth in daily active users (DAUs)[7]. Other well known players include Epic which has also launched its own Unreal Editor for Fortnite (UEFN) earlier this year with a promise to share 40% of revenue[8] with creators who contribute to content creation. However, modding in traditional games faces some limitations. In Minecraft for example, users that mod the game have to create a new server which fragments the ecosystem further since modded versions introduce new barriers to entry and are not interoperable with other modded games.

The case for composable modding is well-backed and fully on-chain games are able to **unleash the potential of the creator-content-player flywheel, enabling players to mod the game without fragmenting game economy and generate strong network effects for the platform.** Under the concept of an "Autonomous World", ecosystems governed by their own set of rules are interoperable, allowing creators and gamers to seamlessly interact across different game environments. Gaming experiences are also set to be more immersive and everlasting in such environments.

Figure 4: Composability in On-Chain Games



Source: Binance Research

A notable example of a composable, fully on-chain game is Realms.World: Eternum. Developed by BibliothecaDAO, Eternum aims to be the foundational infrastructure layer upon which people are able to build and extend the ecosystem. The layer also comes bundled with an early SDK, along with initial game systems, allowing for maximum expansion.

True User Ownership of Digital Assets

Vitalik Buterin, founder of Ethereum, once remarked that the developers of Wars of Warcraft weakened his character against his will[9], which exposed him to the whims of centralized services. Evidently, assets in traditional, centralized games are never truly owned by the players themselves. Despite that, the desire and appeal of asset ownership within the game economy is evident in traditional games. For example, Counter Strike skins which are tradable outside of the game witnessed one rare skin being sold for a whopping US$160,000 in March 2023. World of Warcraft ("WoW") gold was also valued at 7x[10] Venezuela's bolivares which led to a clamp down by Blizzard that negatively impacted players' experience. **A permissionless, fully on-chain game that uses NFTs to transfer ownership rights to gamers can foster more robust interaction among players.** This is due to a deeper sentimental attachment to assets that are under the gamers' full control, effectively reducing the risk of censorship.

Community Governance

The introduction of in-game governance tokens to provide gamers with governance rights, places the future development of the game in the hands of engaged players. Through the formation of DAOs or gaming guilds, it ensures a transparent, frictionless, self governance by a global community powered by blockchain technology that eliminates the reliance on rent-extracting middlemen. This could foster a more close knit community, improve retention rate and lower the cost of acquisition.

One notable case is Star Atlas DAO which empowers gamers with stakes in the development of the game. Using a voting escrow model in which POLIS tokens are staked in exchange for voting power, this model hands the development of the game to the most convicted community members. The Star Atlas community has demonstrated vested interest in the game, introducing various tools and features to expand the ecosystem alongside the core game developer which include:

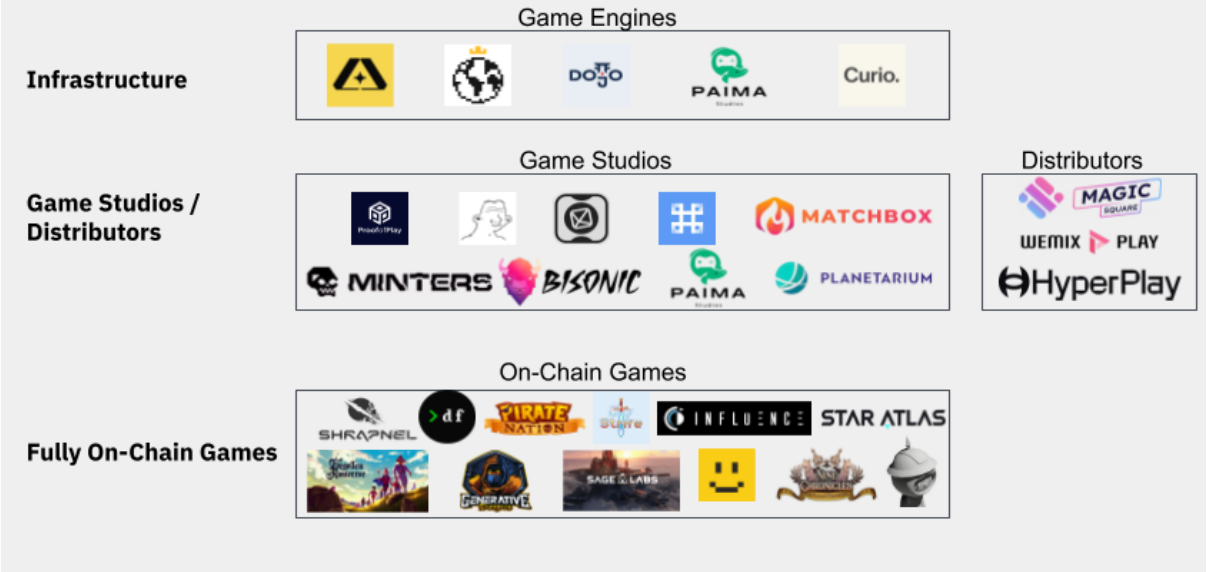
- **Redspatula's Star Atlas Ships Pricing Tool:** data aggregator on Atlas ships
- **The Hologram News Network:** news site dedicated to the Star Atlas's ecosystem
- **Galactic Dash:** Dashboard focusing on Star Atlas ships, politics and economics
- **Atlas Theory:** Video content creation dedicated to Star Atlas

In order to build a self-sustaining, community-driven governance model, the developer has laid out a framework consisting of the DAO, Star Atlas Constitution, Star Atlas Council and the Foundation. Eventually, after evidence of success, the developer also plans to transfer the Emergency Override Authority that is used to protect the DAO and community, into the hands of the DAO[11].

3 On-Chain Gaming Ecosystem

Today's on-chain gaming landscape can be broken down into several categories with each playing a pivotal role in advancing the growth of fully on-chain games.

Figure 5: Landscape of Gaming



Source: Binance Research. Note that this is not an exhaustive representation of the ecosystem.

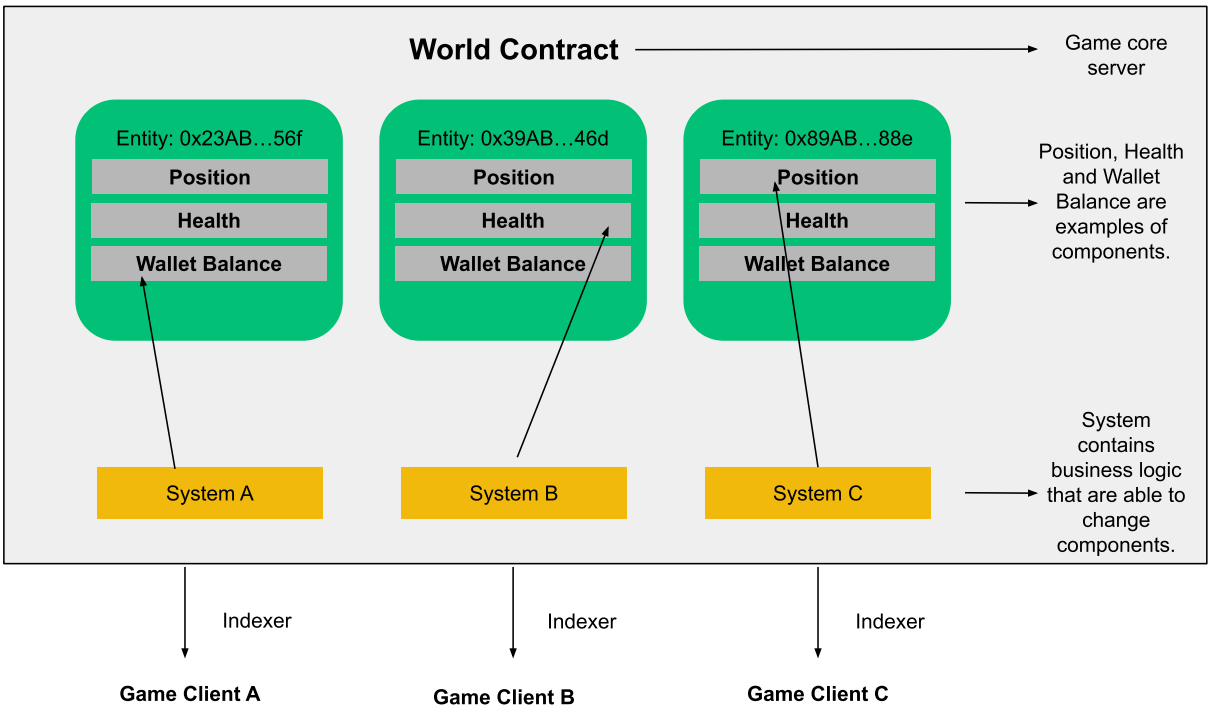
Infrastructure

Positive developments in the area of game engines have paved the way in easing the development of on-chain games. Before we dive into the innovations of some of these notable game engines, it is first important to understand the concept of Entity, Component, System ("ECS") which is widely used in traditional as well as on-chain game development. Essentially, **ECS is a popular gaming framework that defines the interactions, relationships, and operational updates between game objects.** Used by major game engines such as MUD and Dojo, **ECS has been effective in separating data (components) from logic (system) and this vastly improves scalability of games as content update** can be done at the system level without necessitating massive contract redeployment and data loss as a result.

ECS Framework

- **Entity:** refers to unique identifiers / game subjects. Entities are able to add / modify or remove components dynamically through the use of Systems.
- **Component:** refers to features of the game subjects.
- **System:** refers to functions that are able to cause a change in the component of the entity.

Figure 6: ECS Architecture



Source: Binance Research

The battle-tested nature of the ECS framework has garnered mainstream traction and some even came up with variations of ECS framework such as Action Registry Core ("ARC") by Jump Crypto.

Figure 7: On-Chain Game Engines

	MUD	Paima Engine	Dojo	World Engine	Keystone
Year Developed	2022	2022	2023	2023	2023
Blockchain Compatibility	EVM	EVM, Cardano, Polkadot, Astar, Algorand	Starknet	EVM	EVM
Architecture	Tables (Store)	ECS	ECS	ECS	ECS
Composability	Composable	Composable	Composable	Composable at the base shard level	Composable
Provability	Yes, non-native	Mix of native and non-native	Yes, native with Cairo	Yes, non-native	Yes, non-native
Languages	Solidity	Javascript, Solidity	Rust, Cairo	Go	Go
Client Stack Support	React, Phaser, Godot, Unreal, Unity	React, Phaser, HTML5 Canvas, Unity, Game Maker, Vite	React, Bevy Engine, Unity & Unreal	Phaser, Unity, Unreal	React, Unity
Open Source	Yes	Yes	Yes	Yes	Yes
Games Built	OPCraft, Sky Strife, Primodium, Words3, Battle for Blockchain, Aether Sands	Tarochi, Wrath of the Jungle: Tower Defense	Roll Your Own, Influence, Loot Survivor	Dark Frontier	Treaty, Warcraft

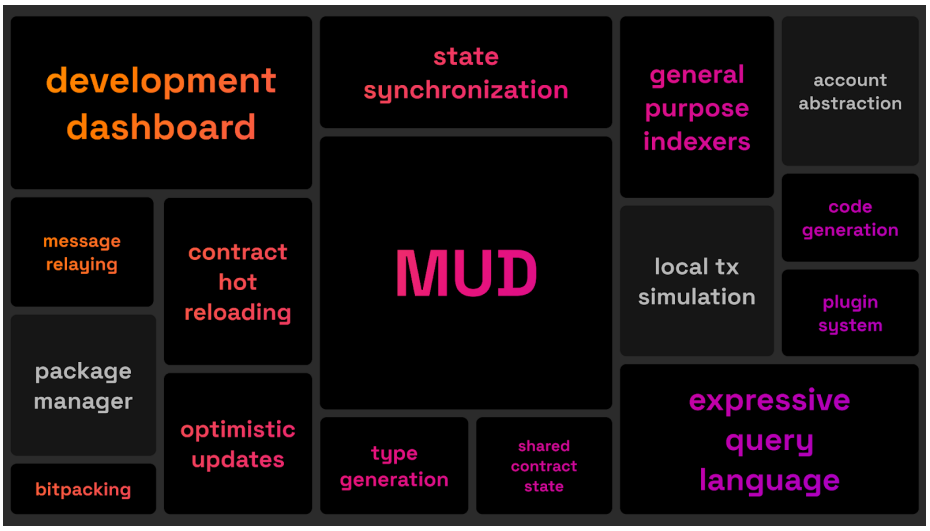
Source: Binance Research

For illustration, we will touch on MUD, Dojo and Paima Engine in more detail in the next few pages. Keystone and World Engine are also notable ecosystem participants. **Keystone** is developed as a **rollup framework built on top of OP Stack** that caters to high-tick rate games and uses Celestia as a DA layer, mitigating costs for users. **World Engine's innovative sharding architecture** aims to provide **geolocation optimization that can further lower game latency** for gamers across the world.

MUD: The Catalyst For On-Chain Gaming Adoption

In the past, on-chain games faced several uphill hurdles such as **difficulty in upgrading logic in smart contracts, lack of synchrony between chain and client leading to inconsistencies in game state, irregularities in access control that made it difficult for third parties to further develop the project**. With that in mind, the MUD engine was developed to address the challenges and ease the development of on-chain games. MUD, developed by Lattice, is a framework used for complex Ethereum applications. It **introduces a standard data model and logic and abstract away the low level complexities.** In April 2023, MUD underwent an upgrade to become MUD v2[12] which introduces a wider functionality.

Figure 8: MUD Engine full range of solutions



Source: MUD, Github

Currently, with **more than 90% of on-chain games using the MUD engine on EVM networks**, MUD has proven itself to be a popular solution to building on-chain games. Several key elements of MUD have played an important role in its adoption.

1. Use of **'Store'**: Store, an embedded EVM database, is an alternative to Solidity's storage engine and prides itself on offering **automatic indexing and packing data more tightly than Solidity's storage engine** by using a custom encoding scheme. This enhances developer experience and reduces the storage costs significantly as compared to using Solidity. New data tables can also be developed at runtime compared to Solidity's static data, enhancing scalability and upgradability.

2. With store being the data storage layer, **'World'** provides the primitives for logic and access control on top of it. World serves as the central entry point and forward calls to systems which are smart contracts that contain the business logic of the game. Indexer provided by MUD listens to the state updates to components within the world contract and communicates to the clients **ensuring contract-client synchronization.** World also **allows permissionless extensions** that facilitate the

development of a suite of tools to improve the developer experience. Some of the plugins that are compatible with the world contract now include Endless Quest, MUDVRF, and MUDSCAN.

3. **Mode.** Mode is an automatic, off-chain indexer that automatically state sync with MUD clients to speed up the initial loading time.

MUD v2 is helpful as an engine to develop on-chain games as it provides seamless **automatic contract upgrades, state synchronization between contract and client** without custom networking code, **optimistic updates** that deliver lower latency time. Compared to v1 which uses ECS architecture, v2 provides support for relational databases, providing developers with the flexibility in deciding their data structures.

Moving forward, MUD promises to elevate the user experience further in future by introducing built-in support for **account abstraction, contract package manager, local simulation of transactions, live client state updates from indexer** to enhance the responsiveness of gameplay.

Dojo: The First Provable Game Engine On Starknet

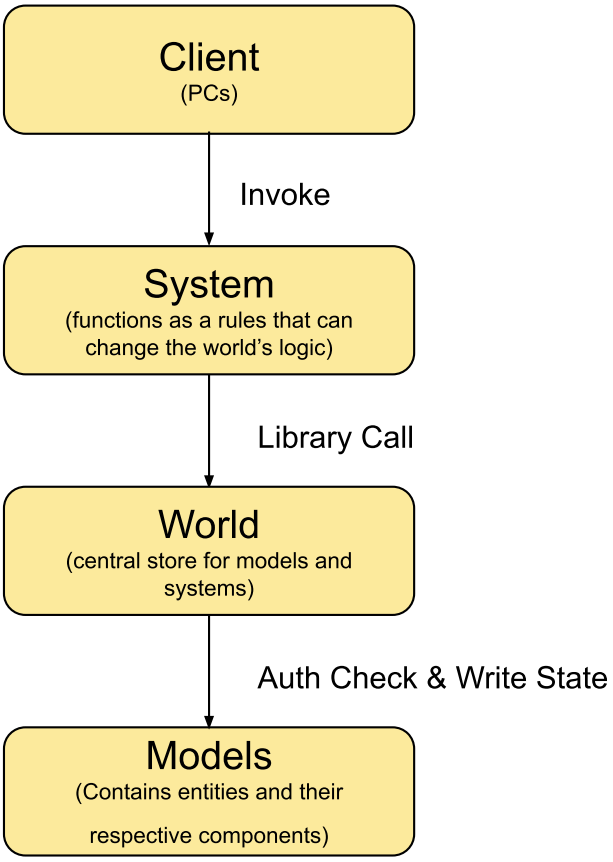
While MUD works across all EVM chains, Dojo[13] is a gaming engine **dedicated to the Starknet ecosystem and is also a popular choice among developers.** Dojo is a Cairo-based game engine that aims to deliver a seamless developer experience with a **developer stack that consists of Torii, Katana, and Sozo.** With a gaming-focused architecture, Dojo facilitates the growth of on-chain RTS, MMO, TCG and RPG games. Some of the games include Realms.World:Loot Survivor, Influence, Roll Your Own and Briq. Below we will explore the key features of Dojo that make it one of the most popular gaming engines for on-chain games.

Cairo is a Turing complete, Rust-like language that uses state differences[14] to describe the state transitions that occurred during the game and posts just the execution output on the Data Availability layer such as Ethereum. Compared with some optimistic rollups which post all the inputs of all intermediate state transitions on the Data Availability layer, it is evident that **Cairo offers a greater verification efficiency and reduces the costs to verify storage proofs. Cairo proofs are also recursive**, which means that an infinite amount of compute can be compressed into a single, constant-sized proof that can be verified in constant time, making it suitable for games with high interaction frequency. Writing apps in Cairo can also bring **privacy through the integration of zero-knowledge proofs and interoperability.** Verification can be broken down into parts with private sensitive information being verified on the client side before on-chain verification. Programs running in Cairo Virtual Machine are compatible not only with Starknet L2, Starknet appchains, but also through browsers via WebAssembly (“WASM”).

Architecture of Dojo ECS Model

Underlying Dojo is its core game architecture, ECS, that paves the way for an interconnected network of worlds, promoting an autonomous world.

Figure 9: An overview of Dojo architecture



Source: Binance Research, Dojo

Toolchains

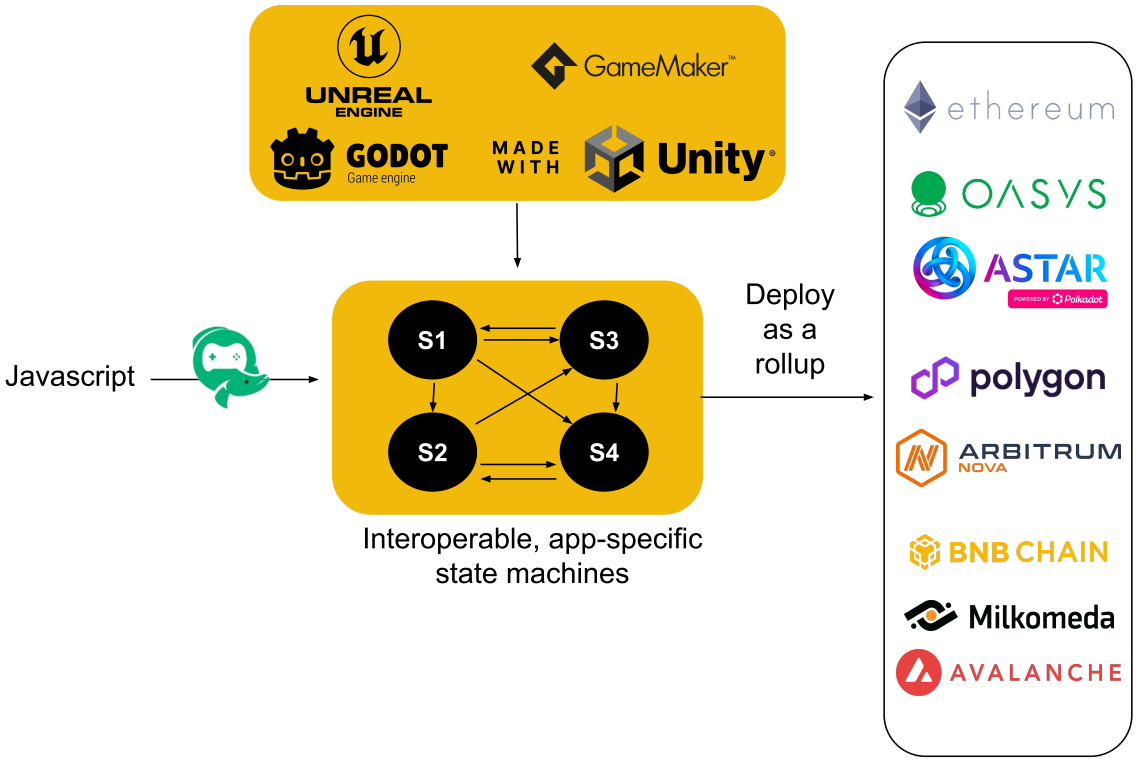
- **Torii:** Building on-chain games often requires dealing with constant indexing of the game state. Torii is an automatic indexer that provides a low latency, high performance GraphQL and gRPC interface for game clients allowing real-time rendering of game state changes. Using Torii drastically reduces the effort to build on-chain games and eliminates the need to build an indexer which itself could be tedious.
- **Katana:** Katana is an extremely fast local node that supports swift iteration of the game logic. Used as the devnet, developers can use Katana to test the transactions that are sent during the game. Katana also supports convenient RPC methods that can easily be used to change network configurations such as block times and transaction fees.

Moving forward, Dojo announced future enhancements that include **L3 game research, optimistic updates, client-side proving and cross chain transfer** that would further pave the way to build a seamless autonomous world.

Paima Engine: Powering the Autonomous World

Paima Engine[15] is a game engine used to create on-chain games and Autonomous Worlds, allowing developers to use programming languages such as Javascript. Developers are able to **use the familiar Javascript language to build their game state machine as L2s sovereign rollups and also leverage on Solidity to build composable parts of the game**, making Paima Engine a flexible developer tool.

Figure 10: Using Paima Engine to build on-chain games as sovereign rollups



Source: Paima Engine, Binance Research

The game engine currently supports EVM chains and also provides support for several non-EVM chains through a novel account abstraction framework. Several key features play a vital role in Paima Engine being the engine for the Autonomous World.

Deployment of games as sovereign rollups

While most games either use optimistic rollups or zk rollups, Paima engine's choice to build games as sovereign rollups has brought some notable benefits including increased scalability as compared to zk and optimistic rollups.

Stateful NFTs and NFT compression

Stateful NFTs are NFTs that are able to capture the evolving game state. The evolving NFTs can also be decoupled into its elements, opening up the possibility of engaging marketplace trading experience. This unlocks the potential of MMOs in the web3 world yet still inheriting the underlying security.

Paima Whirlpool

With Paima Whirlpool, developers are no longer confined to a single blockchain, giving them the ability to deploy their games cross-chains. This increases accessibility and advances the development of an Autonomous World. Embedded within Paima Whirlpool is the Paima Batcher that enables cross-chains play, automatically paying gas fees for users (account abstraction) and decreasing overall fees by batching game inputs together. For example, **the game Jungle Wars built by Paima Engine enables full fledged cross-chain support using Paima Whirlpool.** Users are able to interact with the game regardless of whether they are using Metamask, Polkadot or Cardano Wallet.

Parallelization

The deployed sovereign L2 rollups are not only more efficient than EVM, they also support running state machine updates in parallel, bringing massive scalability to on-chain games although this is currently not natively available in EVM.

Other than these features, Paima Engine's partnerships with other projects also enhances the value for developers. Firstly, a partnership with **ZekoLabs, a zk Layer for Paima** and an ecosystem project on the Mina protocol, **introduces developers to a zk rollup framework.** Paima also partners with **Shinkai Network, which serves as a peer-to-peer AI layer** for the Autonomous Worlds, providing AI agents in games.

Moving forward, the team has stated **plans to incorporate full support for non-EVM chains such as Mina protocol, Cardano, and Polkadot while also adding compatibility with other programming languages.** Storage costs will also be substantially lower in future as the team works towards tapping Celestia and Avail for its DA needs.

On-Chain Games

With the advancement in technology, we are seeing a convergence of builders and gamers into on-chain games. Builders are able to have a more pleasant experience when creating on-chain games while gamers reap the benefits from these games. The alignment of interest has led to the launch of some of the most popular on-chain games recently such as Nine Chronicles and Star Atlas: SAGE Labs.

Figure 11: A comparison among some of the notable on-chain games.

	Dark Forest	Nine Chronicles	Realms.World	Sage Labs	Pirate Nation	Sky Strife
Year Founded	2020	2020	2021	2023	2023	2023
Developer	0xParc	Planetarium	BibliothecaDAO	ATMTA	Proof of Play	Lattice
Categories	Combat-based RTS	MMORPG	Fantasy Metaverse	RPG, Strategy	RPG, Strategy	Combat-based RTS
Chain	Ethereum & Gnosis	Ethereum & BNB	Starknet	Solana	Ethereum & Arbitrum Nova	Redstone Testnet
Gaming Engine	MUD	Libplanet	Dojo	Unreal Engine 5	Developed internally	MUD
Key Features	• Uses zk-SNARK to facilitate gameplay yet preserving privacy • Mini games held on Altlayer for scalability • Players engage in quests to compete for resources and conquer planets	• Powered by $NCG • PvP, PvE battles to earn resources that can be used in-game • Uses both Odin and Heimdall blockchains • Available on Google play and Apple store	• A fantasy multiverse that contains fully on-chain games like Eternum, Loot Survivors • Leverages on Dojo tech suite • Has a standalone loot module that enables developers to integrate resources across different games	• Powered by the ATLAS token and an extension of the Star Atlas economy • Dynamic resource management • Advanced fleet dynamics	• Dual currencies model (PGLD, MARK) that can be used to level up and craft items • Gasless, signless transactions • In-built VRF feature	• Permissionless composability. Players are able to create their own matches • Season Pass offers holders access to exclusive maps, free matches, rewards, access list among others

Source: Binance Research, Project websites

Nine Chronicles



Source: Nine Chronicles

Nine Chronicles is the world's first, open-sourced, decentralized, MMORPG game powered by the Libplanet engine and developed by Planetarium, its parent company. Nine Chronicles has witnessed early success as it saw more than 200,000 unique users pre-registered for the game. Gamers who pre-registered are able to claim rewards after downloading the app.

DappRadar has also listed it as one of the top dApps in November in terms of unique active wallets. The game has also risen through the ranks to become the top 5 games in Japan. On November 22, the game announced the highly anticipated release of its game in mobile version which provides exposure to the world's 1.7 billion mobile gamers serving as a tailwind for further growth. With active collaboration such as the likes of SuperWalk, a move-to-earn game, Nine Chronicles looks to continue its strong growth momentum.

Notable partners of Nine Chronicles include Binance Labs, Animoca Brands, Ubisoft, Sandbox, Sfermion among others.

Background and Gameplay

Nine Chronicles is an online RPG game set in a fantasy world where players can engage in quests, competitive PvP, PvE events and earn $NCG rewards as well as other resources. Resources earned can be used to upgrade existing equipment, craft new equipment and level up character.



Source: Nine Chronicles Mobile Version

Game Features

- Game ecosystem is made up of 9 different realms with 50 stages each
- The underlying token powering the economy is the $NCG which can be earned through participation in events, quests and campaigns. $NCG can be used to buy resources to craft powerful items, trade with other players, challenge other players and accelerate growth.
- Assets values on the games are determined by the free market.
- Staking: Monster Collection, allows players to stake $NCG for a one-month period and be rewarded with Hourglasses and AP potions on a weekly basis.
- Currently, there is no entry fee allowing players to easily join the game and gamers are able to play the game using familiar logins such as Google without having to create a web3 wallet.

SAGE Labs



Source: SAGE Labs

Launched on 21 September, SAGE Labs is a fully on-chain game on the Solana blockchain that extends the Star Atlas economy, following the early success of its predecessor SAGE: Escape Velocity which recorded more than 80 million transactions. SAGE Labs is currently in early access and is a precursor to its upcoming game SAGE: Starbased that promises on a complete 3D graphics that delivers an expanded immersive gameplay.

Notable partners supporting Star Atlas include Animoca Brands, Big Brain Holdings, Republic Capital, Mechanism Capital among others.

Background and Gameplay

SAGE Labs offers a unique gameplay experience as it incorporates crypto elements such as tokenized asset management with traditional gaming features like resource extraction and fleet management. In the 2D browser experience, players create fleets to explore the space economy, extract resources, craft golden tickets to have the opportunity of winning Star Atlas loots which include ships, lands, ATLAS tokens among others in a prize pool that is worth US$1.35 million. Success in the game is determined by the composition of their fleets which is vital in this galactic exploration game.

As an extension to the Star Atlas economy, SAGE Labs brings large-scale utility to the native token ATLAS and its circular economy. Players are able to use current items interchangeably with current and future game releases as well.

Game Features

- Uses the Solana blockchain to record and verify in-game transactions ensuring security and transparency.
- Real time mining algorithm that simulates real-time and real-live mining
- Advanced fleet dynamics that allow players to have the ability to create, manage and disband fleets, offering a strategic depth unseen in other games.
- SAGE Labs will encompass 51 sectors or systems and all of Star Atlas NFTs will be usable in some aspects of the game
- In-game marketplace that uses the ATLAS token as the trading currency

4 The Hurdles

With less than 10% of blockchain games being fully on-chain games, it is evident that there is still a long way to go. The sector has several challenges to address before fully on-chain games gain mainstream adoption.

Technical Constraints Inhibiting Innovative Game Design

Currently, most games exist on EVM chains, but EVM is still not optimal for the deployment of certain types of on-chain games, often resulting in subpar player experience and limited engagement.

Solidity, being the dominant programming language on EVM chains, is also not conducive for the development of complex games as it is not scalable and it often generates costly fees especially for games that come with a large number of transactions. The storage of **large amounts of gaming-related data also adds additional costs.** In traditional video games, the most common programming languages are C++, Javascript which are favorable for creating games since they are general purpose languages, have high computational performance and a faster time-to-market than Solidity[16]. Solidity, being a programming language tailored for the deployment of smart contracts, also introduces a barrier to entry for developers who are more comfortable with traditional programming languages.

Blockchain fundamental design is also different, with state updates every block time. Game state updates, however, are executed via a tick-based system. **A tick is a snapshot in time recording all the properties of all objects in the game world.** Tickrate refers to the number of times the game state is being updated and broadcasted every second. As a result, this might introduce a higher latency rate which might not be optimal for certain games such as the likes of FPS, MMORPG or high tickrate games. Fortunately, we are already seeing healthy developments by **Argus Labs and Curio in introducing tickrate to the chain level to address this limitation.**

User Experience

In order to onboard the masses, improvements in user experience should be developers' key priority. The need to create a web3 wallet, load it with the blockchain's native gas tokens and submit transaction approval for every action introduces an added layer of complexity and inconvenience that prevent gamers from having a seamless gaming experience, leading to low retention rate consequently. However, as technology advances with the rollout of **account abstraction (AA) technology such as Cartridge Controller by Cartridge that enables a one-time pre-approval of applications for a seamless experience**, poor UI/UX would eventually be a thing of the past.

Fragmented Gaming Ecosystem

Today's on-chain gaming still remains **fragmented as players are unable to bring their digital assets outside of its ecosystem into other realms of on-chain gaming.** Players are compelled to use different wallets for different games leading to inconvenience. This limits the potential synergy that could potentially be extracted if these games were interoperable with each other, think the ability to move digital assets from between games of the same category. However, we are seeing progress as MUD, an on-chain gaming engine, provides a universal interface through the use of standardized data model that allows anyone to easily deploy system smart contracts to implement interoperable behavior.

AI Bots

On-chain games can be overwhelmed with bots as players seek to cheat the system. **Therefore, it remains a challenge for on-chain games to differentiate between bots and real players and rewarding only active players in the game.** Developers need to build games that are protected against bot metas and sybil collusion to prevent negative gaming experience among genuine gamers. Fortunately, some games such as Shrapnel are showing promising progress in this area. Shrapnel is partnering with Anybrain to build models that are able to detect fraudulent behavior such as the use of aimbots.

5 Catalysts for Onchain Games

With an increasing number of traditional video game publishers making their entry into on-chain gaming, it wouldn't be surprising to see a wider range of tech stacks supporting the growth of various gaming categories on-chain.

From a fundraising perspective, it is also clear that there is huge interest in supporting the growth of on-chain games. For example, SHRAPNEL, a fully on-chain AAA competitive, first-person shooter game, has raised US$20 million in a Series A fundraising round led by Polychain Capital. Proof of Play has also raised a substantial amount of US$33 million in funding from investors such as a16z, Anchorage Digital among others. Evidently, the future of interoperable gaming metaverse holds great promise for the flourishing gaming industry.

As the industry forges towards greater adoption of on-chain games, we believe that there are several catalysts that could propel the growth of the industry.

Catalyst 1: Engaging Gameplay with sustainable design mechanics

While not a fully on-chain game, we have seen how the play-to earn model spurred the rapid growth of Axie Infinity, which has at one point recorded notable daily unique active wallets ("UAWs") of more than 728K but now declined to 17.24K[17] as of 1 December according to DappRadar. The peak of the play-to-earn model has faded during the bear market and ultimately developers should focus on building an experience-based game by supporting elements like user-generated content, ownership permanence and sustainable tokenomics.

An engaging gameplay experience should encourage a healthy growth in user-generated content as this will not only help reduce development fees from the developers' perspective but also create a more engaged audience which could lead to a higher retention rate. According to Frederick Reichheld of Bain & Company[18], a 5% increase in retention rate could lead to significant increase in bottom line ranging from 25% - 95%. A prime example of this would be Dark Forest which is also one of the pioneers in fully on-chain games that is launched in 2020. With the ability to mod the existing game, its community of more than 10,000 players have come around to organize competitions within the game, built marketplaces, tools that automate game functions and even bots to play the game themselves, resulting in a self-sufficient, constantly evolving gameplay experience that has lasted through the changing market environment.

A sustainable game design should be one that prudently manages token emission and rewards the active players. This is essential as on-chain gaming continues to deal with the issue of bots within the gameplay which reduces the incentives for other genuine gamers. Gameplay experience should also be fair for all players since that is one of the main

appeals of having a fully on-chain game. Hence, we would likely see more adoption of technology such as Chainlink VRF that help introduce fairness and randomness in games.

Catalyst 2: More investment into infrastructure technology to enable frictionless onboarding

Current adoption for on-chain gaming is low due to factors such as having to create a eb3 wallet, purchase native gas tokens, steep learning curve and poor UI/UX. Getting the next leg up in adoption requires more than just having interesting gameplay. **Without a seamless gaming experience found in traditional gaming, the barrier to entry continues to deter new users from using on-chain games.** Infrastructure providers like Web3Auth that enable fuss-free wallet creation using Twitter or Google OAuth coupled with payment gateway solutions by service providers such as thirdweb are positive catalysts.

Account abstraction would likely see more innovations and investment since they are critical in easing the entry into on-chain gaming. Proof of Play creates a burner wallet for gamers on Pirate Nation, which is custodial and used to grant permissions to Proof of Play to pay for the users' transactions, all the while still ensuring the security of their assets. Other AA technology likely to see more traction include bundled transactions and social recovery.

While ECS is currently the dominant architecture used to deploy games, that could change as new frameworks are introduced to improve game development. One example of this would be the ARC framework that was introduced by Jump Crypto to facilitate the development of fully on-chain games on the Solana network.

Instead of relying on EVM smart contract for on-chain logic and storage of game state, we could see new alternatives gaining traction such as Molecular Execution Machine ("MEM") which is a serverless functions network that is chain agnostic, supports a wide range of languages, and is highly scalable with near-instant finality. MEM also has integration with Arweave to enable a permanent game state storage[19].

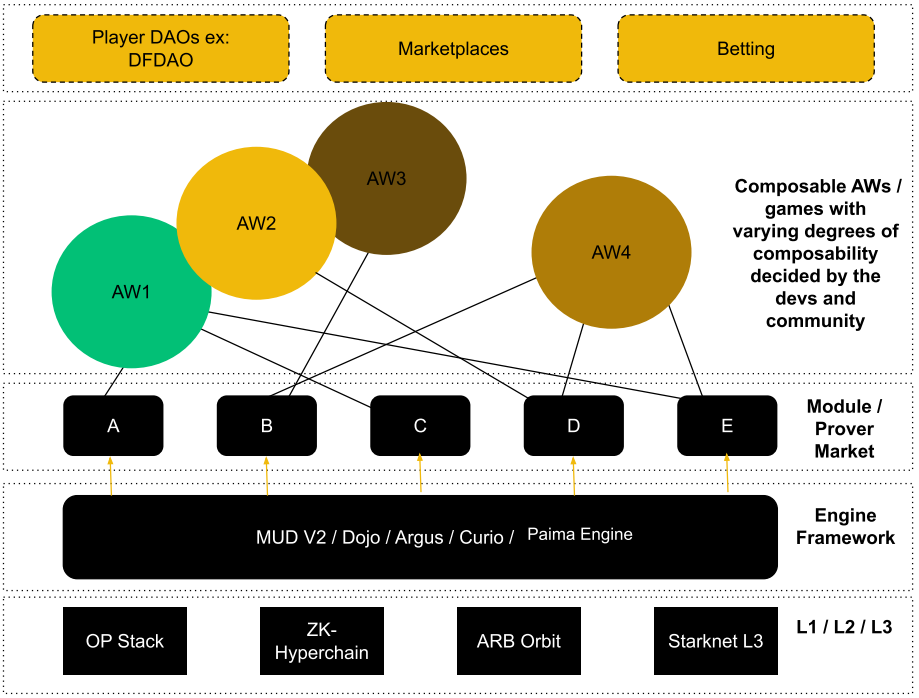
We could also see an increasing focus on L3s leveraging on the concept of fractal scaling on zk-rollups to further enhance scalability of games, accommodating a wider range of games. Fractal scaling, a concept developed by StarkWare, refers to the growth of app-specific chains built recursively over L2s, boasting hyper-scalability and better control of tech stack while retaining security from the underlying L1.

Catalyst 3: Interoperable Economies (Autonomous Worlds)

We believe that interoperable game economies also known as "Autonomous Worlds" will be the next catalyst for on-chain game adoption. **Autonomous Worlds promise to usher in a more immersive gaming experience as gamers shift from the mentality of just achieving their goals and exiting the game to being an integral part of an ever-evolving virtual world.** The formation of an autonomous world also formalized interaction rules removing the need for privileged individuals to keep the world alive providing a

permissionless, trustless, permanence guarantee. We are already seeing exciting developments in this area with Realms.World building an interconnected network of composable on-chain games where players are able to bring their assets from one game to another.

Figure 12: Ecosystem of an Autonomous World



Source: IOSG Ventures, Binance Research

6 Closing Thoughts

Fully on-chain games are still in their nascent stage with lots of potential to grow driven by catalysts such as technological advancements and adoption by traditional game studios among others. Although we will likely not see hybrid and traditional games entirely usurped by fully on-chain games in the near future, the deep integration of blockchain technology into fully on-chain games paints a promising future for an Autonomous World that could deliver a more immersive gaming experience, alongside other benefits mentioned in this report.

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